Lesson notes · for you

Samples and uncertainty

Data science · Level 4 · Lesson 1

Time
30 min

Materials
A notebook and a pencil; 50 cards or paper slips numbered 1 to 50, and a bag; A phone or tablet with Darsi, for the lab

Learning goals

- See why samples vary.
- Report a margin of error.
- Ask who is missing.

Activity: Cards in a bag

1. Warm-up (5 min): Two surveys found 50 and 60 minutes of homework. Say who you think is right.
2. Draw (10 min): With a partner, draw samples of 5 and of 20 cards and write each average.
3. Compare (10 min): Plot both sets of averages and say which crowd closer together.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Why are the averages different?
- Which samples wobbled less?
- Who could a school survey miss?

Check yourself

- Write a survey result as a range with a margin of error.

Try it this week
Put numbered cards in a bag, draw ten at a time and work out each average. Why are your averages all a little different?

Worksheet

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Name _____ Class _____ Date _____

1 **We want a smaller margin of error. What should we do?**

◯ Ask more pupils, chosen at random ◯ Round the answer ◯ Ask only our friends

2 **Making a biased sample larger removes the bias.**

◯ True ◯ False

3 **Compute the standard error s / √n and print the 95% confidence interval: mean ± 2 SE, rounded to 1 decimal place.**

```
mean, s, n = 55.0, 12.0, 36
se = 0
lo = round(mean - 2 * se, 1)
hi = round(mean + 2 * se, 1)
print(lo, hi)
```

Write your program:

4 **Scale the sample.**

To halve the standard error, multiply n by _____. To cut it to a third, multiply n by _____.

Word bank: 2 · 4 · 9 · 3

5 **An app-based poll of 10,000 users finds 80% support. What limits its validity? Choose all that apply.**

☐ Users chose whether to answer ☐ The sample is too small ☐ Non-users are excluded

✎ **Think of a survey or poll you have seen. Who might have been missing from it?**

Check your answers

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1 **Ask more pupils, chosen at random**

A bigger random sample wobbles less.

2 **False**

Size reduces random error only; the systematic gap remains.

3

```
mean, s, n = 55.0, 12.0, 36
se = s / n ** 0.5
lo = round(mean - 2 * se, 1)
hi = round(mean + 2 * se, 1)
print(lo, hi)
```

SE = 2, so the interval is 55 ± 4, from 51 to 59 minutes.

4 **To halve the standard error, multiply n by 4. To cut it to a third, multiply n by 9.**

SE falls with √n: √4 = 2 and √9 = 3.

5 **Non-users are excluded; Users chose whether to answer**

The problem is selection and non-response bias, not size.